DANIEL OMONIGUHNO OKENE

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RESEARCH INTEREST

Computational Fluid Dynamics, Multiphase Flow Modelling, Turbulence Modelling, Mixing and Transport Phenomena, Numerical Methods for Fluid Mechanics, Scientific Computing, and High-Performance Computing.

EDUCATION

University of Benin, Benin City, Edo State, Nigeria **01/2020 – 04/2025**

Bachelor of Engineering (BEng), Mechanical Engineering

Final Grade: First Class Honors (**4.69/5.0** || **3.828/4.0**; Scholaro) **Class Rank:** 1 out of 109.

Thesis Title: Computational Fluid Dynamics (CFD) Analysis of Cavitation Effects on Centrifugal Pump Performance and Flow Patterns in Oil and Gas Applications Using ANSYS.

RESEARCH EXPERIENCE

Research Assistant **05/2024 – 05/2025**

Thermal-Fluid Lab, The University of Benin *Supervisor: Dr. Ikechukwu Owunna*

- Conducted a CFD investigation of cavitation and performance degradation in centrifugal pumps using ANSYS Fluent, combining the Volume of Fluid multiphase model, RNG k-ε turbulence model, and Schnerr–Sauer cavitation model.
- Investigated the effects of inlet pressure, wall roughness, and backflow conditions on cavitation development, internal flow behaviour, and pump performance.
- Performed model validation and numerical sensitivity studies to assess solution reliability, developing practical experience in multiphase flow simulation, turbulence modelling, numerical convergence, and CFD validation.

Research Collaborator **04/2026 – Present**

Next Energy Xystems Technologies (NEXT) Research Hub *Supervisor: Prof. Ighodaro*

- Conducting off-design thermodynamic modelling and performance assessment of a combined-cycle power plant under baseload and part-load operating conditions.
- Evaluating component-level energy and exergy performance to investigate the effects of changing operating conditions on system efficiency and thermal losses.

Summer Research Intern **06/2024 – 08/2024**

The European Organization for Nuclear Research [CERN] – Geneva, Switzerland

- Collaborated with CERN and University of Naples Federico II researchers on structural-dynamics modelling of the SHiP experiment's decay vessel, developing the vessel geometry in ANSYS SpaceClaim and reproducing a published wave-guided modal-analysis methodology using ANSYS Mechanical and MATLAB.
- Computed modal characteristics and dispersion curves and applied the methodology to the full vessel geometry, gaining experience in numerical modelling, computational validation, and research workflows involving high-performance computing. [Internship Report]

Research Assistant **09/2024 – 03/2025**

Materials and Ballistic Laboratory, The University of Benin *Principal Investigator: Prof. Ighodaro*

- Developed and validated kenaf fiber-reinforced composite properties and models using the two-step homogenization techniques and experimental test data to capture composite behavior for impact simulation.
- Simulated high-velocity ballistic impacts on kenaf fiber-reinforced composites using Abaqus Explicit, focusing on stress distribution, deformation, and energy absorption.

COMPUTATIONAL PROJECTS

2D Lid Driven Cavity Flow Solver [Repo]

- Developed a C++ finite-difference solver for the incompressible Navier–Stokes equations using the pressure projection algorithm. Implemented iterative Poisson solvers, residual convergence monitoring, and post-processing routines for comparison with the Ghia benchmark.

D2Q9 LBM Channel Flow Solver [Repo]

- Developed a C++ D2Q9 Lattice Boltzmann solver for incompressible channel flow. Implemented the BGK collision model, streaming procedure, and bounce-back wall treatment, and validated the numerical solution against the analytical Poiseuille velocity profile.

PUBLICATION

- Owunna, I., **Okene, D.**, Madu, C., Nwabueze, B. (2025). A Numerical Approach to Studying Crude Oil Cavitation in Centrifugal Pumps. ***National Institute of Professional Engineers and Scientists 3rd International Conference.***
- Ighodaro, O., Egware, H., Nwankwo, E., Orumwense, E., Kwasi-Effah, C., Abdulazeez, M., Amune, J., Okpako, O., Esimike, T., **Okene, D.**, Madu, C., Dan-Odhomo, J., Ogunmola, B. (2025). Multiscale Homogenisation and Quasi-static Characterisation of Kenaf 2D Woven Fabric Composites: Foundational Insights for Future Ballistic Modelling. *Progress in Engineering Science, Elsevier.* ***https://doi.org/10.1016/j.pes.2025.100191***
- Ighodaro O., Oluwole, A., **Okene, D.**, Madu, C., (2026). Off Design Modelling and Performance Assessment of The Afam VI Integrated Combined Cycle Power Plant. *Applied Energy (***Elsevier), In Review.**

CONFERENCES AND POSTER PRESENTATION

- Presented my research work titled: "Investigation on the Structural Dynamics of the Decay Vessel for the SHiP Experiment" at the CERN summer student poster session. **26th July 2024**. [Poster]
- "A Numerical Approach to Studying Cavitation in the Oil and Gas Industry" Presented at the National Institute of Professional Engineers and Scientists 3rd International Conference. Included in the book of abstracts. University of Benin, **9-11 April 2025**.

TECHNICAL SKILLS

- **Computational:** Programming (Python, C, C++, Git, Bash), High-Performance Computing (Linux, Slurm, CPU/GPU Parallelisation), Scientific Computing (SciPy, NumPy, Matplotlib, Pandas, PyTorch)
- **Software:** Fusion 360, SOLIDWORKS, ANSYS (Mechanical, CFX, Fluent), OpenFOAM, and Microsoft Suites.

PROFESSIONAL EXPERIENCE

Finite Element Software Developer, Africa Makes – (Remote) **08/2024 – 03/2026**

- Taught the theoretical foundations of FEA to software developers, ensuring accurate implementation of numerical methods in solver development. Contributed to the development of Africa's first cloud-based FEA solver (Enitan), validated numerical outputs and served as a QA tester for the Enitan Studio and Classroom applications, conducting functionality testing, and providing technical feedback to enhance user experience and solver accuracy.

Facilities Engineer Intern, Chevron Nigeria Limited **01/2024 – 05/2024**

- Monitored the reconstruction of tank farms in OKAN under the supervision of a project engineer. Learned and applied knowledge of P&ID and process engineering to support project activities.
- Participated in Process Hazard Analysis (PHA) and Hazard and Operability (HAZOP) sessions to ensure operational safety. Attended budget defense meetings with the National Upstream Investment Management Services (NUIMS).

TEACHING EXPERIENCE

Graduate Assistant, Department of Mechanical Engineering, University of Benin. **05/2025 – 11/2025**

- Assisted in teaching, preparing course material, grading tests, and exams for Fluid Mechanics III – V (MEE461, MEE561, and MEE562). Additionally, I anchored tutorials and advised final-year undergraduates on their projects.

Academic Tutor, University of Benin **02/2023 – 04/2025**

- Organized weekly tutorials on Engineering Mechanics, Thermodynamics, and Fluid Mechanics to over 100 undergraduate students across several departments, contributing to a 70% improvement in test and exam performance.

COMPETITIONS

NADDC, Maiden Campus Shuttle Electric Vehicle Design Competition **04/2025 – 07/2025**

- Member of the Vehicle Body and Structural Design Team, where I carried out research on EV skateboard chassis and performed aerodynamic CFD simulations to determine the best vehicle body configuration from our box fish model.

Autodesk Designathon 2024 **06/2024**

- Member of Team Chevy, where I modelled and designed a waste collector vehicle that operates both on land and water using Fusion 360. [Certificate of Participation]

VOLUNTEERING AND LEADERSHIP EXPERIENCE

Volunteer, Bedrock Human Development Foundation **08/2022 – Present**

- Coordinated the logistics team for the 9th UNIBEN Marathon Competitions and the second edition of the academic scholarship examination, ensuring smooth participant movement and materials distribution.

Director of Academics, Nigerian Institute of Mechanical Engineers, UNIBEN Chapter. **03/2023 – 09/2023**

- Collaborated with over 10 scholars across different levels to organize tutorials for 150+ students, significantly boosting departmental academic performance. Initiated and coordinated scholarship and career development webinars, enhancing students' access to opportunities [Students won over **$6000** in scholarships].

Team Lead and Brand Designer, AIESEC in Benin **03/2021 – 04/2022**

- Led a team of 15 designers in curating templates and assets for campaigns and events, strengthening the Local Committee's visual identity. Directed designs for Finance Week and YALI Collaboration Events, contributing to broader regional visibility and partner engagement.

HONOURS AND AWARDS

- Black in AI, Emerging Leader in AI Program Recipient **07/2026**
- ISI Graduate Study Scholarship Recipient **06/2026**
- Best Graduating Student, Department of Mechanical Engineering, University of Benin **04/2025**
- Best Undergraduate Final-Year Research Project, Department of Mechanical Engineering, UNIBEN **04/2025**
- MTN Science and Technology Scholarship Recipient **09/2023**
- The Youth Center Academic Scholarship Recipient **06/2023**
- Midwestern Oil and Gas Company Scholarship Recipient **02/2022**
- Bedrock Foundation Scholarship Recipient **07/2022**
- 2X John Asiruwa Excellence Award for Best 200 & 300 Level Mechanical Engineering Student **05/2023**
- Award of Selfless Leadership – Christ Apostolic Church Youth Fellowship UNIBEN/UBTH **02/2025**
- Award of Recognition – Nigerian Institute of Mechanical Engineers. UNIBEN Student Chapter **10/2020**

TRAININGS AND CERTIFICATIONS

- **OpenFOAM on HPC Systems** | *Austrian Scientific Computing (ASC) Research Center* **Ongoing**
- **Artificial Intelligence Bootcamp** | *TechCrush* **07/2026 - 10/2026**
- **Turbulence Modelling** | *IIT Madras* **07/2025 - 10/2025**
- **Computational Fluid Dynamics Using Finite Volume Method** | *IIT Madras* **07/2025 - 10/2025**
- **Introduction to High Performance and Parallel Computing** | *University of Colorado Boulder* **05/2025 - 07/2025**
- **Machine Learning Specialization** | *DeepLearning.AI X Stanford University* **10/2024 - 04/2025**
- **Aspire Leaders Program** | *Aspire Institute and Harvard Business School* **08/2025 - 10/2025**
- **Fundamentals of Manuscripts Preparation** | *Elsevier Research Academy* **09/2025**
- **SolidWorks CAD Design Associate (CSWA)** | *Dassault Systèmes* **09/2024**
- **Introduction to Ansys Fluent** | Ansys **01/2024 - 02/2024**
- **Computer Aided Engineering** | *University of Abuja – HEPSSA –Royal Academy of Engineering* **03/2023 - 03/2024**

PROFESSIONAL MEMBERSHIPS

- CERN Alumni Network.
- Nigeria Institute of Mechanical Engineers, NiMechE.
- Graduate Member, Nigeria Society of Engineers, NSE
- National Institute of Professional Engineers and Scientists, NIPES.